Double Exponential Distribution

1 Definition

A continuous random variable X is said to have a Double Exponential Distribution / Laplace distribution if it has the PDF as,

$$f(x) = \frac{1}{2\sigma} e^{-\left|\frac{x-\mu}{\sigma}\right|} \quad x \in \mathbb{R}, \mu \in \mathbb{R}, \sigma > 0.$$

2 Median and Mode

$$f_x(\mu - h) = f_x(\mu + h), \forall h$$

$\Rightarrow$ The distribution is symmetric about μ and the median is μ. Also, the mode is at μ.

3 Moments

$$\begin{aligned} E|X-u|^r &= \int_{-\infty}^{\infty} |x-u|^r \frac{1}{2\sigma} e^{-|\frac{x-u}{\sigma}|} dx \quad \frac{x-\mu}{\sigma} = z \\ &= \int_{-\infty}^{\infty} \sigma^r |z|^r \frac{e^{-|z|}}{2\sigma} \sigma dz \qquad \frac{dx}{\sigma} = dz \\ &= \sigma^r \int_0^{\infty} e^z z^r dz \\ &= \sigma^r \Gamma(r+1) \end{aligned}$$

which is a convergent Gamma integral if $\overline{r+1} > 0$

$$\begin{aligned} &\Rightarrow r > -1 \\ &\rightarrow r \geqslant 0 \end{aligned}$$

$\therefore \mu_r'$ exists if $r \varepsilon \mathbb{N}$.

Expectation also exist. As the distribution is symmetric about μ,

$$E(X) = \mu$$

$\mu_{2r-1} = 0$ [All odd order central moments are zero].

$$\begin{aligned} \mu_{2r} = E(X-\mu)^{2r} &= \int_{-\infty}^{\infty} (x-\mu)^{2r} \frac{1}{2\sigma} e^{-|\frac{x-\mu}{\sigma}|} dx \\ &= 2\int_0^{\infty} \sigma^{2r} z^{2r} \frac{1}{2\sigma} e^{-z} dz \\ &= \sigma^{2r}\Gamma(2r+1) = \sigma^{2r}(2r)! \text{ if } r \text{ is an integer.} \end{aligned}$$

$E(X-\mu) = \int_{-\infty}^{\infty} (x-\mu) \frac{1}{2\sigma} e^{-|\frac{x-\mu}{\sigma}|} dx = \int_{-\infty}^{\infty} \sigma z \frac{1}{2\sigma} e^{-|z|} \sigma dz = 0$

$$\therefore E(X) = \mu$$

4 CDF

$$f_X(x) = \begin{cases} \frac{1}{2\sigma} e^{-\frac{\mu - x}{\sigma}} & x < \mu \\ \frac{1}{2\sigma} e^{-\frac{x-\mu}{\sigma}} & x > \mu \end{cases}$$

$F_X(x) = \int_{-\infty}^{x} f_X(t)dt$

For $x < \mu$, $\quad F_X(x) = \int_{-\infty}^{x} \frac{1}{2\sigma} e^{\frac{t-\mu}{\sigma}} dt = \frac{1}{2\sigma}\sigma e^{\frac{x-\mu}{\sigma}} = \frac{1}{2} e^{\frac{x-\mu}{\sigma}}$

For $x > \mu$,

$$\begin{aligned} F_X(x) &= F_X(\mu) + \int_{\mu}^{x} \frac{1}{2\sigma} e^{\left(\frac{t-\mu}{\sigma}\right)} dt \\ &= \frac{1}{2} - \frac{1}{2} e^{-\frac{x-\mu}{\sigma}} + \frac{1}{2} \\ &= 1 - \frac{1}{2} e^{-\frac{x-\mu}{\sigma}} \end{aligned}$$

$$F(x) = \left\{ \begin{array}{ll} \frac{1}{2} e^{\frac{x-\mu}{\sigma}} & \text{if } x \leqslant \mu \\ 1 - \frac{1}{2} e^{-\left(\frac{x-\mu}{\sigma}\right)} & \text{if } x > \mu \end{array} \right\}$$

5 Problems

If $X \sim DE(\mu, \sigma)$, find the distribution of $|Y - \mu|$.

$$\begin{aligned} F_Y(y) &= P[Y \leqslant y] \\ &= P[|X - \mu| \leqslant y] = P[-y \leqslant X - \mu \leqslant y] \\ &= P[\mu - y \leqslant X \leqslant \mu + y] \\ &= F_X(\mu + y) - F_X(\mu - y) \end{aligned}$$

PDF of Y,

$$\begin{aligned} \frac{d}{dy} f_Y(y) &= f_X(w + y) - f_X(w - y) \\ &= \frac{1}{2\sigma} e^{-y/\sigma} + \frac{1}{2\sigma} e^{-y/\sigma} \\ &= \frac{1}{\sigma} e^{-y/\sigma} \end{aligned}$$

Remark

In a prediction problem our model is,

$$y_i = f(x_i) + e_i \quad \forall i = 1(1)n$$

If $e_i \sim N\left(0, \sigma^2\right)$, the PDF of $(e_1 \dots e_n) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n e^{-\frac{1}{2}\sigma^2 \sum_{i=1}^{n} e^2}$

Otherwise, if $e_i \overset{\text{iid}}{\sim} \text{DE}(0, 1)$, the PDF of $(e_1 \dots e_n)$ is given by,

$$\left(\frac{1}{2\sigma}\right)^n e^{-\frac{1}{\sigma}} \sum_{i=1}^{n} |e_i|$$

Assume e_i's are most probable then the best prediction is available if we minimize $\sum |e_i|$. Therefore Normal law and DE law are respectively known as Laplace 1st law and 2nd Law of Error.

Standard Double Exponential Distribution / Standard Laplace distribution

If $X \sim$ Laplace $(\mu = 0, \sigma = 1)$, the distribution is called Standard Laplace distribution.